

Integrating Reinforcement Learning for Behavioral Analysis in Interconnected Electricity and Carbon Markets

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Abstract—The integration of electricity and carbon markets plays a crucial role in promoting renewable energy adoption and achieving global carbon reduction targets. However, the interconnection of these markets introduces significant challenges due to their inherent complexity and dynamic nature. This paper proposes a hybrid experimental learning framework, combining real-world experimentation and reinforcement learning-based generative models, to analyze participant behavior and market dynamics in coupled electricity and carbon markets. The framework simulates trading processes by modeling strategic bidding behaviors using a reinforcement learning approach, formulated as a Markov Decision Process (MDP). Leveraging both historical and experimental data, the model generates bidding strategies that adapt to market conditions while balancing economic and environmental objectives. A case study based on the Guangdong electricity market, incorporating modified IEEE 6-bus systems and real-world carbon trading scenarios, demonstrates the effectiveness of this approach in capturing market complexities and uncovering distinct behavioral patterns among participants. Results highlight the ability of reinforcement learning-based generative models to replicate human decision-making and optimize bidding strategies in interconnected markets. The findings provide insights into market behaviors, regulatory impacts, and potential strategies for improving efficiency and sustainability in the energy sector. This study underscores the value of hybrid experimental learning as a tool for exploring coupled market dynamics and guiding the transition to low-carbon energy systems.

Index Terms—Electricity Market, Carbon Market, Deep Learning, Generative Models

I. INTRODUCTION

The relationship between electricity and carbon markets is integral to global efforts aimed at reducing carbon emissions and advancing renewable energy adoption. The interaction between these markets creates significant opportunities for integrated policy-making, which has the potential to expedite the transition to sustainable energy systems[1]. Nonetheless, managing the complexities associated with these markets poses considerable challenges, requiring innovative strategies for decision-making and operational efficiency.

Both markets play critical roles in tackling environmental and sustainability issues. The electricity market drives renewable energy adoption by fostering a competitive environment where different energy sources compete based

on cost and efficiency[2]. It also influences the equilibrium of supply and demand, while shaping decisions related to investment and technological progress in energy generation and distribution[3]. Meanwhile, the carbon market operates as a regulatory and trading mechanism, primarily through cap-and-trade systems that set emission limits and permit the trading of carbon allowances. This framework incentivizes emission reductions by creating economic benefits for low-carbon initiatives[4]. The convergence of these markets facilitates the formulation of policies that integrate energy production and environmental sustainability, encouraging a comprehensive approach to achieving long-term sustainability goals.

Engaging with the interplay of these markets introduces distinctive strategic complexities. The electricity market, with its competitive bidding systems and price volatility, demands constant adaptability from participants[5] [6]. Conversely, the carbon market requires navigating not only trading activities but also adherence to regulatory obligations, compelling participants to align financial goals with environmental commitments[7].

The joint operation of these markets amplifies these complexities, creating an environment where integrated strategies must account for the intricate dynamics of both markets. This calls for a deep understanding of market behavior and regulatory structures[8]. The challenges in these combined markets are inherently multifaceted, involving economic, environmental, and regulatory dimensions[9].

To address the intricacies of joint electricity and carbon markets, this paper proposes an innovative methodology that merges real-world experimentation with computational simulations. This approach enables the modeling and analysis of market behaviors in a controlled yet dynamic setting[10]. By leveraging this methodology, researchers can gain valuable insights into market dynamics and participant behaviors that are otherwise difficult to capture through traditional analytical methods.

II. METHODOLOGY

This section outlines a framework describing how entities interact and trade within the electricity and carbon markets. In the carbon market, participants decide on their buying or

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selling quotas and set corresponding prices. Meanwhile, in the electricity market, they establish bidding curves for the spot market, defined by power and price pairs. The system incorporates the interdependent dynamics between these two markets, recognizing that behaviors in one market can significantly impact the other. By integrating these behavioral inputs, the simulation system generates trading outcomes, offering a platform for analyzing market interactions and uncovering key insights into their dynamics.

To capture and model the social behaviors within these markets, a hybrid experimental learning approach is employed. This method functions as a scenario simulation tool, collecting decision-making data in a controlled setting to study participant actions. At its core, this approach relies on a generative model trained on data from these experimental setups to simulate potential scenarios and behaviors. This comprehensive framework facilitates an in-depth understanding of decision-making processes, providing valuable insights into how market participants adapt and interact in complex environments.

A. Hybrid Experiment

Generative models play a crucial role in hybrid experiments by replicating and analyzing human behavior. These models effectively capture participant behavioral patterns using experimental data to simulate social interactions. By leveraging a data-driven approach, generative models provide valuable insights into complex decision-making processes and participant dynamics.

Hybrid experiments are conducted in multiple rounds to collect sufficient data for training generative models. Depending on the experimental setup, these models can be trained either in real-time (online) or after the data collection phase (offline). Once trained, generative models can replace some human participants in subsequent experiments, enabling extended simulations while utilizing insights derived from the initial human interactions.

Generative models bring several benefits to economic experiments. They significantly reduce exploration costs by simulating diverse scenarios without requiring extensive human involvement. Furthermore, these models help standardize decision-making processes by providing participants with guided, professional frameworks. As a result, they enhance the consistency and reliability of experimental outcomes, leading to more robust analyses.

The generative model operates as an inverse probability density function, which typically lacks a clear mathematical representation due to the non-interpretability of many machine learning techniques. In cases where interpretability is desired, a probability estimator can be employed after the generative model is trained. This approach enables the creation of an interpretable probabilistic model that provides analytical insights into participant behavior. However, when the primary goal of the hybrid experiment is market simulation, interpretability may not be essential, and the focus can remain on the generative model's simulation capabilities.

B. Simulation Process

The simulation process for the experiment follows a structured flow, starting with the initialization of market entity parameters and decision-making models. The simulation timeline then begins, during which decision drivers are computed using kernel density estimation—a critical step that shapes the behavior of market participants. Based on these drivers, both human participants and computer agents make decisions, resulting in bilateral matchmaking and clearance of trades.

As the simulation progresses, it evaluates whether the final time point has been reached. If not, the process continues to the next time point, ensuring a dynamic and iterative approach. Once the final time point is reached, the simulation concludes. This systematic framework enables a detailed analysis of bidding behaviors within interconnected electricity and carbon markets, providing insights into their complex dynamics.

III. GENERATIVE MODELS

This study investigates the utilization of reinforcement learning-based generative models as an efficient method for modeling strategic offering behaviors in the coupled electricity and carbon markets. These models are designed to simulate complex decision-making processes by formulating the strategic bidding problem as a Markov Decision Process (MDP), enabling adaptive learning of optimal strategies through interactions with a simulated environment.

The core of the reinforcement learning framework consists of an agent that observes the state of the market, selects actions (bidding strategies), and receives rewards based on the outcomes of those actions. The state space includes variables such as electricity prices, carbon prices, market quotas, and generator-specific constraints. The action space consists of strategic decisions, such as electricity bidding and carbon quota transactions. The reward function is designed to incentivize behaviors that align with both economic and environmental objectives, such as profit maximization, emission reductions, and compliance with regulatory constraints.

The agent's policy is trained using advanced reinforcement learning algorithms, such as Proximal Policy Optimization (PPO) or Deep Q-Networks (DQN). These algorithms iteratively optimize the agent's policy by balancing exploration and exploitation, ensuring convergence to optimal strategies over multiple episodes of interaction with the environment.

The objective of the reinforcement learning model can be expressed as follows:

$$\pi^*(s) = \arg \max_{\pi} \mathbb{E} \left[\sum_{t=0}^T \gamma^t r_t | \pi \right] \quad (1)$$

Where: - π represents the policy that maps states to actions.
- s is the current state of the market environment.
- r_t is the reward received at time t .
- γ is the discount factor that determines the importance of future rewards.

In our application to the electricity and carbon markets, reinforcement learning enables the model to capture and adapt to dynamic market conditions, generating strategies that are

robust and reflective of real-world complexities. The trained policy effectively generalizes across diverse market scenarios, providing practical insights into participant behaviors and market dynamics.

The reinforcement learning-based approach leverages neural networks to approximate the policy and value functions, enabling it to handle the high-dimensional state and action spaces inherent in coupled markets. The policy network maps the observed market state to a probability distribution over possible actions, while the value network estimates the expected cumulative reward for a given state. Additionally, advanced feature engineering techniques are employed to preprocess market data and enhance the network's ability to identify latent patterns in the input space, improving the quality of the learned policies.

The training process involves iterative updates to the policy and value networks, guided by feedback from the environment. The agent explores various bidding strategies, gradually refining its policy to achieve optimal outcomes. During training, techniques such as reward shaping and adaptive learning rates are applied to accelerate convergence and stabilize learning. This framework provides a flexible and data-driven approach to modeling strategic behaviors, offering significant advantages in terms of adaptability, scalability, and interpretability for decision-makers in coupled markets.

Moreover, the proposed model incorporates mechanisms to handle market uncertainties, such as price volatility and fluctuating carbon allowances, by integrating stochastic elements into the state representation. This enables the agent to learn robust strategies that can adapt to unexpected market changes, ensuring its applicability in real-world scenarios. The results demonstrate that the reinforcement learning framework effectively balances economic objectives with environmental constraints, supporting market participants in navigating the complexities of coupled electricity and carbon markets.

IV. SIMPLE CASE STUDY IN THE COUPLED ELECTRICITY AND CARBON MARKET

A. The Electricity System Setting

A modified IEEE-6 bus systems is used in this chapter as shown in Figure to show the results of the hybrid experimental learning. The thermal units are at bus 1, 2 and 3. The loads are at bus 4, 5, and 6. The wind power is injected at bus 2 and 3.

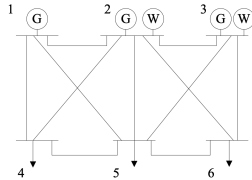


Fig. 1. Topology for the modified IEEE 6-bus system

The experimental setup focuses on a simulated electricity grid comprising six buses and five generators. The daily

load on the grid is modeled using a load forecast error that aligns with a standard normal distribution at each time point. This methodology simulates the daily variations in electricity demand realistically, accounting for the unpredictable nature of consumer usage patterns.

The five generators involved in the setup are managed by human participants, as detailed in Table I of the paper. These participants are responsible for controlling the output of the generators, providing a hands-on element to the simulation. This setup allows for the analysis of human decision-making in response to changing grid conditions, contributing valuable insights into how operators might manage similar real-world systems.

TABLE I
PARAMETER OF THE 5 GENERATORS IN THE IEEE-6 BUS SYSTEM

No.	Location	Capacity (MW)	Fixed cost (Y/h)	Variable cost (Y/MWh)
1	Bus 1	610	8100	325
2	Bus 2	780	9809	289
3	Bus 3	680	9240	297
4	Bus 2	240	480	0
5	Bus 3	360	720	0

B. Electricity and Carbon Market Setting

In this study, Guangdong electricity market was used as the simulation market. The wholesale electricity market in Guangdong comprises two primary components: a long-term transaction market and a spot market. The spot market is divided into day-ahead and real-time markets, where trading and pricing are adjusted to reflect the immediate supply and demand conditions. The hybrid experiment in this study focuses on simulating the day-ahead market with a consistent expected load, aligned with the spot market rules in Guangdong.

In the day-ahead market, the bidding strategy is represented by a five-stage price-power supply curve, $S(p, q)$. This curve outlines five stages of ascending prices and their corresponding power segments. The prices are constrained within the market's upper and lower bounds, set at ¥1000/MW and ¥0/MW, respectively. This structure ensures that the bidding strategy remains realistic and adheres to the regulatory framework.

The five-stage bidding strategy remains consistent throughout the day, providing a stable framework for participants to engage in the market. Additionally, the constraints related to long-term transaction volumes are removed to more accurately reflect real-world conditions, allowing for a more flexible and responsive bidding environment.

For each type of generator, the default carbon emission quota factor σ is set at 0.8 t/(MW · h). The carbon emission intensity for thermal power generators is set at 0.89 t/(MW · h), whereas wind generators have an emission intensity of 0 t/(MW · h). At the start of the experiment, all generators receive their carbon allowance based on their historical power generation levels. In addition, to simplify the carbon market

simulation, we assume that one entity just needs to make one decision every day about whether to buy or sell the carbon emission quota and the corresponding price and quantity.

C. Setting of RL-Based Generative Models

The case study utilizes reinforcement learning (RL)-based generative models to generate bidding strategies using data from human participants and historical records. The training data for the RL agent is categorized into "simulated" data from human participants and "historical" data from actual bidding records.

The simulated data is collected from professional traders, undergraduates, and graduate students engaged in 52 rounds of hybrid experiments. Meanwhile, the historical data is sourced from the actual bidding records of specific generators in the Guangdong pilot spot market.

To enhance the training data, the Mixup data augmentation method is employed to create a 4:1 mix of simulated and historical data for each generator in the basic scenario. In the experimental scenarios where no historical data is available, the bootstrap method is used to augment the simulated data[11].

The RL-based generative framework models the bidding strategy generation process as a Markov Decision Process (MDP). The state space includes variables such as electricity prices, carbon prices, and generator-specific constraints. The action space consists of possible bidding strategies, represented as multi-stage price-power supply curves. The reward function is designed to incentivize economically and environmentally optimal bidding behavior, balancing profit maximization with carbon emission constraints.

The RL agent's policy is trained using Proximal Policy Optimization (PPO), a widely used RL algorithm that ensures stable and efficient learning. PPO optimizes the following objective:

$$J(\theta) = \mathbb{E}_t \left[\min(r_t(\theta) \hat{A}_t, \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_t) \right] \quad (2)$$

Where: - $r_t(\theta)$ is the probability ratio of the new policy to the old policy. - $\hat{A}_t$ is the advantage function. - ϵ is a hyperparameter that controls the clipping range.

The policy network is parameterized using a neural network with multiple hidden layers. The input to the network includes features extracted from market conditions and generator-specific attributes. The output represents the probability distribution over possible bidding strategies. The value network, which estimates the expected cumulative reward for a given state, shares similar architecture and is trained alongside the policy network.

The training process involves iterative interactions with the simulated environment, where the agent explores various strategies and learns to optimize its policy through reward feedback. The policy is periodically updated to improve performance, ensuring convergence to robust bidding strategies across diverse market conditions.

To account for the potential achievement of mixed Nash equilibrium, we establish the total number of training epochs at 10,000. We adopt binary cross-entropy (BCE) as our preferred measure for the real/fake classification, observing convergence of the GANs at varying epochs, with the BCE either approaching zero or stabilizing in fluctuation for both the generator (G) and discriminator (D). For illustrative purposes, a single converged example from one of the 50 GANs is depicted in Fig. 2.



Fig. 2. Generator 1 GAN Training Convergence

D. RL Agent Performance

This study utilizes 10 reinforcement learning agents to model the bidding strategies of five generators across two different cases. Each RL agent is trained using 10,000 episodes, with initial data derived from a combination of 52 groups of experimental data and 25 historical records in a 4:1 ratio. For Cases 2, which lack historical records, all training data is augmented from experimental data.

TABLE II
BIDDING OF RL AGENT FOR GENERATOR 1

	Case 1		Case 2	
	Price	Quantity	Price	Quantity
Stage 1	0.95	5.22	0	0
Stage 2	298.19	482.18	299.13	482.92
Stage 3	310.62	98.29	312.18	29.82
Stage 4	829.89	12.4	899.23	19.02
Stage 5	989.21	40.1	989.21	0
Carbon market	60.21	+4000	60.18	+4080

Example results for Generators 1 and 4 highlight distinct bidding behaviors between thermal and wind generators, as shown in Table 2.2 and Table 2.3. In these two tables, the first five rows represent the behavior in the electricity market, and the last row represents the behavior in the carbon market. The typical behavior of coal generators involves allocating most of their output to the second and third price stages while reserving a smaller portion for the fifth stage. Conversely, wind generators tend to allocate nearly all their power to the second stage.

Notably, electricity prices for the second and third stages for coal generators are higher in Cases 2 compared to Case 1, while electricity prices for the fifth stage for wind generators

TABLE III
BIDDING OF RL AGENT FOR GENERATOR 4

	Case 1		Case 2	
	Price	Quantity	Price	Quantity
Stage 1	0	0	0	0
Stage 2	163.39	198.23	164.49	197.57
Stage 3	237.63	30.23	237.63	29.87
Stage 4	382.47	0	382.47	0
Stage 5	786.21	0	786.21	0
Carbon market	60.54	-310	60.57	-300

are lower. These outcomes demonstrate the unique bidding behaviors of different types of generators under various scenarios in the coupled electricity and carbon markets.

V. CONCLUSION

This paper presents a comprehensive hybrid experimental learning framework for analyzing and modeling participant behaviors in coupled electricity and carbon markets. By integrating real-world experimentation and computational modeling, including reinforcement learning-based generative models, this approach addresses the complexities of interconnected market dynamics and provides valuable insights into strategic decision-making under varying market conditions.

The proposed methodology effectively captures the synergies and trade-offs between the electricity and carbon markets, enabling a nuanced understanding of participant behaviors, regulatory impacts, and market dynamics. The reinforcement learning framework demonstrated the ability to model strategic bidding behaviors, leveraging Markov Decision Processes and advanced learning algorithms to adapt to evolving market environments. The case study based on the Guangdong electricity market and the IEEE 6-bus system validated the framework's capability to simulate realistic scenarios, revealing distinct behavioral patterns between different types of generators under various settings.

This study highlights the significance of integrating computational and experimental approaches for analyzing coupled market dynamics. It provides policymakers, market operators, and researchers with a robust tool for developing strategies to enhance market efficiency, support renewable energy integration, and achieve sustainability goals. The results also underscore the importance of incorporating both economic and environmental objectives in designing market mechanisms.

In conclusion, the hybrid experimental learning framework represents a significant advancement in understanding and optimizing behavior in interconnected energy markets. It offers a scalable and adaptable solution for addressing the challenges of managing these markets in a rapidly evolving energy landscape, paving the way for more resilient and sustainable energy systems. Future work could explore extending the framework to include additional market features, such as real-time market feedback, or applying it to other regions and energy systems to further validate and expand its applicability.

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